

Xihui Wu

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Research Interests

Multimodal foundation models, video understanding, compositional video generation, and retrieval-augmented generation.

Education

Shenzhen University

Shenzhen, China

B.S. in Automation, College of Mechatronics and Control Engineering

Sep. 2022 – Jun. 2026

- Selected coursework: Android Application Development(4.5/4.5), Human-computer interface design(4.0/4.5), Object-Oriented Programming and Windows Programming(4.0/4.5), Pattern Recognition(4.0/4.5), C Programming(4.5/4.5).

Paper Under Review

ReCam-VL: Reversal-Aware VLMs for Camera Motion Understanding

Manuscript under review at the AAAI Conference on Artificial Intelligence (AAAI-27), 2026.

Research Experience

Research Assistant

Jun. 2026 – Present

The Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

ReCam-VL: Reversal-Aware VLMs for Camera Motion Understanding

Jun. 2026 – Present

- Co-developed a reversal-closed benchmark of 1,926 real and synthetic videos spanning 14 directional camera motions, with pair-level diagnostics that separate temporal sensitivity from appearance-based shortcuts.
- Contributed to a reversal-coupled Pair-GRPO objective for Qwen3-VL-8B, increasing inverse consistency from 21.0% to 44.2% and reducing unchanged predictions from 35.5% to 14.4%, while improving matched forward/reverse accuracy to 37.1%/31.7%.

Causal-OPD: Counterfactual Branch Credit Assignment for On-Policy Distillation

2026

Working paper.

- Investigating teacher-decision selection for on-policy distillation by scoring matched student and teacher continuations from the same prefix with a task verifier.

MoTIE: Motion Tool-Augmented Inference with Evidence for Camera-Motion Understanding

2026

Working paper.

- Investigating tool-augmented camera-motion understanding by providing frozen VLMs with optical-flow, frame-difference, drift, scale, and parallax evidence through a structured Motion Blueprint.

Research Assistant

Nov. 2023 – Jun. 2024

Institute of Human Factors Engineering, Shenzhen University

Shenzhen, China

- Designed a controlled study of gap-acceptance decisions by novice and experienced electric-vehicle riders under varying traffic-flow conditions.
- Built calibrated virtual-intersection scenarios and participant-facing data-collection software to support repeatable behavioral experiments.

Industry Experience

AI Development Intern

Jul. 2025 – Jun. 2026

深圳流昇智能有限公司

Shenzhen, China

AscendFlowOS: AI Desktop and System Services

Go, Vue, SSE

- Developed desktop and settings modules with configurable layouts, search, application shortcuts, and system widgets; persisted user preferences and connected the desktop to a dedicated AI workspace.
- Built a Go/Echo model gateway with chat, streaming, model-listing, and configuration APIs; integrated conversation history and image inputs, exposing both direct-service and gateway-prefixed routes.
- Implemented Vue chat interfaces with image previews, Markdown rendering, and incremental SSE updates; handled partial stream chunks and disabled server-side response buffering to display output as it arrives.

NexusFlow: Visual AI Workflow Orchestration

React, Node.js

- Implemented a React Flow editor and browser-side graph executor for conditional branches, knowledge retrieval, templated LLM prompts, and HTTP tools; exposed node status and streamed intermediate outputs for debugging.
- Developed document ingestion, structure-aware chunking, embedding generation, and configurable vector-keyword hybrid retrieval for PDF, DOCX, spreadsheet, and text inputs, supplying retrieved context to AI workflows.

- Built a paired-device Node.js runtime with outbound job polling and execution traces; added allowlisted file access and manifest-defined application adapters with per-capability approvals and revocable grants.

AgenticVideoRAG: Evidence-Driven Long-Video QA

Python, Multimodal RAG

- Implemented an iterative planner that decomposes video questions, tracks missing evidence, and issues targeted retrieval queries for entity tracking, state changes, causal questions, and global summaries.
- Built evidence-graph memory linking clips, entities, events, and temporal relations; retained retrieval scores, query provenance, and stepwise traces to make multi-hop reasoning inspectable.
- Added evidence-sufficiency checks and targeted retrieval repair before answer construction; combined semantic graph ranking with temporal-context expansion to collect evidence across distant video segments.

大城智享: WeChat Volunteer and Campus Services

Vue 3, Java, MySQL

- Developed a uni-app/Vue 3 and TypeScript mini program for activity discovery, time-slot registration, cancellation, and profile completion, with Pinia-managed login and user state.
- Built Vue 3/Element Plus administration panels for activity publishing, registration approval and rejection, and CSV exports, connecting participant workflows with organization operations.
- Built Spring Boot REST APIs with MyBatis-Plus/MySQL persistence, WeChat login, JWT validation, database-backed token revocation, and refresh-token rotation.
- Implemented transactional registration with conditional capacity updates and duplicate-enrollment checks; connected approval, attendance, rated checkout, service-hour records, and points rewards.

VUG-RAG: Graph-Based Long-Video Retrieval

- Co-developed a hierarchical Video Understanding Graph for long-video RAG, improving Top-1 accuracy by 12.7% on Entity-VU and reducing inference time by approximately 40%; contributed to graph construction, retrieval, evaluation, and a full internal manuscript draft.

Embedded Systems Intern

Jun. 2025 – Jul. 2025

深圳科创学院

Shenzhen, China

- Developed ESP32 music interaction modules integrating Bluetooth MIDI, matrix touch sensing, and rotary controls with GarageBand.
- Designed erhu pitch/timbre mapping over Bluetooth MIDI and implemented touchscreen playback on ESP32 displays.

Embedded Systems Intern

Mar. 2025 – Jun. 2025

深圳锐盟半导体有限公司

Shenzhen, China

- Contributed to robot motor and IMU/ToF/infrared sensor integration, driver adaptation, and calibration; built LCD expressive eyes and an OLED battery-status display.
- Integrated an LLM-enabled IoT control pipeline with emotion-driven robot expressions, reducing end-to-end response latency from **3 s** to **1 s**.

Selected Work

- **ClawWork:** Built a React Native/NestJS mobile AI workspace with OpenClaw integration and document handling.
- **ValuesRAG:** Built a value-aware knowledge retrieval prototype.
- **AiEditor:** Built a ReAct video-editing agent with tool orchestration and timeline synchronization.
- **ESP32-WeChat:** Enabled Wi-Fi setup and agent management via WeChat.
- **AI-Audio-Hardware:** Built an embedded audio prototype with AI-agent integration.

Technical Skills

Programming: Python, C++, TypeScript

Machine Learning: PyTorch, multimodal language models, reinforcement learning, retrieval-augmented generation

Frameworks & Tools: LangGraph, LangChain, OpenCV, FastAPI, React, Git, Docker, CMake

Languages: English (CET-6)

Honors & Awards

- National-level Undergraduate Innovation and Entrepreneurship Project, 2024.
- Second Prize, East China Regional Competition, National College Student IoT Design Competition, 2024.
- First Prize, East China Regional Competition, National College Student IoT Design Competition, 2023.
- Second Prize, South China Regional Competition, National College Student iCAN Innovation and Entrepreneurship Competition, 2023.